



General Chemistry I

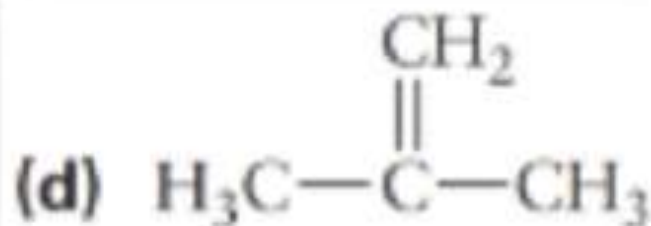
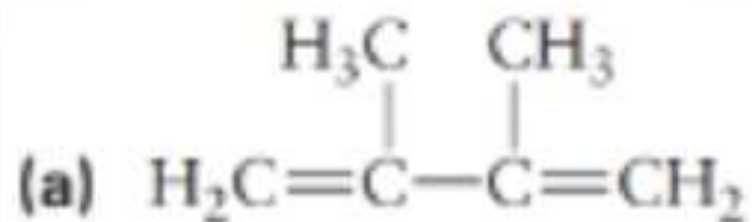
Recitation 10

Siyuan Chen
Dec. 1st

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18		
															Pnictogens	Chalcogens	Halogens			
1	1 H Hydrogen 1.008	Atomic Symbol Name Weight C Solid Hg Liquid H Gas Rf Unknown Metals Alkali metals Alkaline earth metals Lanthanoids Actinoids Transition metals Post-transition metals Metalloids Nonmetals Reactive nonmetals Noble gases																2 He Helium 4.0026		
2	3 Li Lithium 6.94	4 Be Beryllium 9.0122													5 B Boron 10.81	6 C Carbon 12.011	7 N Nitrogen 14.007	8 O Oxygen 15.999	9 F Fluorine 18.998	10 Ne Neon 20.180
3	11 Na Sodium 22.990	12 Mg Magnesium 24.305													13 Al Aluminium 26.982	14 Si Silicon 28.085	15 P Phosphorus 30.974	16 S Sulfur 32.06	17 Cl Chlorine 35.45	18 Ar Argon 39.948
4	19 K Potassium 39.098	20 Ca Calcium 40.078	21 Sc Scandium 44.956	22 Ti Titanium 47.867	23 V Vanadium 50.942	24 Cr Chromium 51.996	25 Mn Manganese 54.938	26 Fe Iron 55.845	27 Co Cobalt 58.933	28 Ni Nickel 58.693	29 Cu Copper 63.546	30 Zn Zinc 65.38	31 Ga Gallium 69.723	32 Ge Germanium 72.630	33 As Arsenic 74.922	34 Se Selenium 78.971	35 Br Bromine 79.904	36 Kr Krypton 83.798		
5	37 Rb Rubidium 85.468	38 Sr Strontium 87.62	39 Y Yttrium 88.906	40 Zr Zirconium 91.224	41 Nb Niobium 92.906	42 Mo Molybdenum 95.95	43 Tc Technetium (98)	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.91	46 Pd Palladium 106.42	47 Ag Silver 107.87	48 Cd Cadmium 112.41	49 In Indium 114.82	50 Sn Tin 118.71	51 Sb Antimony 121.76	52 Te Tellurium 127.60	53 I Iodine 126.90	54 Xe Xenon 131.29		
6	55 Cs Caesium 132.91	56 Ba Barium 137.33	57–71 89–103	72 Hf Hafnium 178.49	73 Ta Tantalum 180.95	74 W Tungsten 183.84	75 Re Rhenium 186.21	76 Os Osmium 190.23	77 Ir Iridium 192.22	78 Pt Platinum 195.08	79 Au Gold 196.97	80 Hg Mercury 200.59	81 Tl Thallium 204.38	82 Pb Lead 207.2	83 Bi Bismuth 208.98	84 Po Polonium (209)	85 At Astatine (210)	86 Rn Radon (222)		
7	87 Fr Francium (223)	88 Ra Radium (226)		104 Rf Rutherfordium (267)	105 Db Dubnium (268)	106 Sg Seaborgium (269)	107 Bh Bohrium (270)	108 Hs Hassium (277)	109 Mt Meitnerium (278)	110 Ds Darmstadtium (281)	111 Rg Roentgenium (282)	112 Cn Copernicium (285)	113 Nh Nihonium (286)	114 Fl Flerovium (289)	115 Mc Moscovium (290)	116 Lv Livermorium (293)	117 Ts Tennessine (294)	118 Og Oganesson (294)		
For elements with no stable isotopes, the mass number of the isotope with the longest half-life is in parentheses.																				
			6 7	57 La Lanthanum 138.91	58 Ce Cerium 140.12	59 Pr Praseodymium 140.91	60 Nd Neodymium 144.24	61 Pm Promethium (145)	62 Sm Samarium 150.36	63 Eu Europium 151.96	64 Gd Gadolinium 157.25	65 Tb Terbium 158.93	66 Dy Dysprosium 162.50	67 Ho Holmium 164.93	68 Er Erbium 167.26	69 Tm Thulium 168.93	70 Yb Ytterbium 173.05	71 Lu Lutetium 174.97		
				89 Ac Actinium (227)	90 Th Thorium 232.04	91 Pa Protactinium 231.04	92 U Uranium 238.03	93 Np Neptunium (237)	94 Pu Plutonium (244)	95 Am Americium (243)	96 Cm Curium (247)	97 Bk Berkelium (247)	98 Cf Californium (251)	99 Es Einsteinium (252)	100 Fm Fermium (257)	101 Md Mendelevium (258)	102 No Nobelium (259)	103 Lr Lawrencium (266)		

Quiz

1. Write structural formulas for cis-4-octene and trans-4-octene.
2. Name the following hydrocarbons:



Homework5

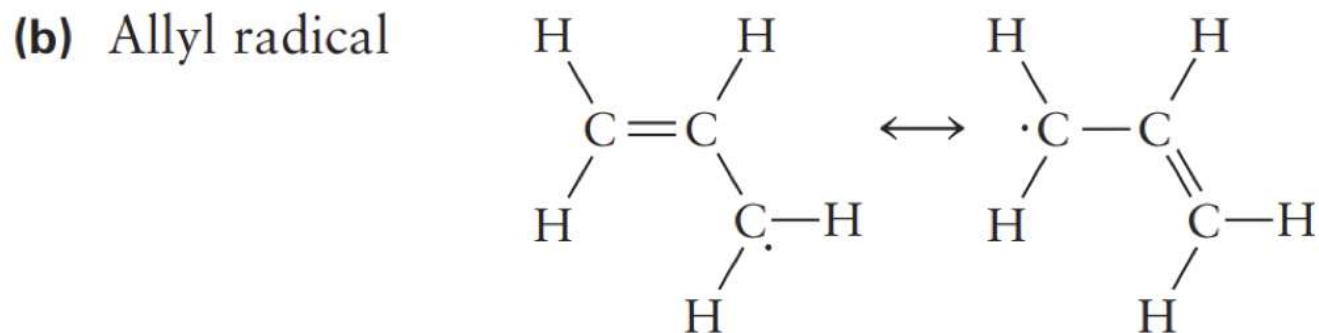
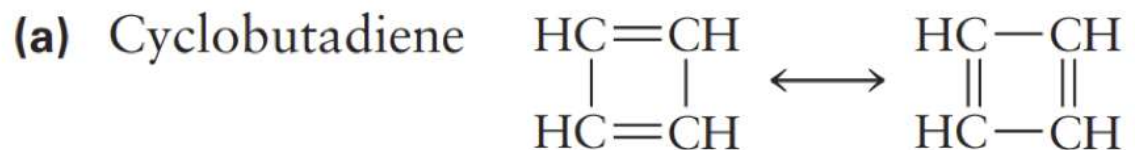
1. Describe the bonding in the bent molecule OF_2 . Predict its energy level diagram and electron configuration.
2. Discuss the nature of the bonding in the nitrate ion (NO_3^-). Draw the possible Lewis resonance diagrams for this ion. Use the VSEPR theory to determine the steric number, the hybridization of the central N atom, and the geometry of the ion. Show how the use of resonance structures can be avoided by introducing a de-localized π MO. What bond order is predicted by the MO model for the N-O bonds in the nitrate ion?

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Homework5

3. For each of the following molecules, construct the π MOs from the $2p_z$ atomic orbitals perpendicular to the plane of the carbon atoms.



Indicate which, if any, of these orbitals have identical energies from symmetry considerations. Show the number of electrons occupying each π MO in the ground state, and indicate whether either or both of the molecules are paramagnetic. Assume that the C atoms in the allyl radical are all sp^2 hybridized.

Homework5

4. (a) Write an equation involving structural formulas for the catalytic cracking of 2,2,3,4,5,5-hexamethylhexane. Assume that the cracking occurs between carbon atoms 3 and 4.
- (b) Draw and name one other isomer of the alkene.
5. Write structural formulas for the following:
- (a) 2,3-Dimethyl-1-cyclobutene
 - (b) 2-Methyl-2-butene
 - (c) 2-Methyl-1,3-butadiene
 - (d) 2,3-Dimethyl-3-ethylhexane
 - (e) 4,5-Diethyloctane
 - (f) Cyclooctene
 - (g) Propadiene
 - (h) 2-pentyne

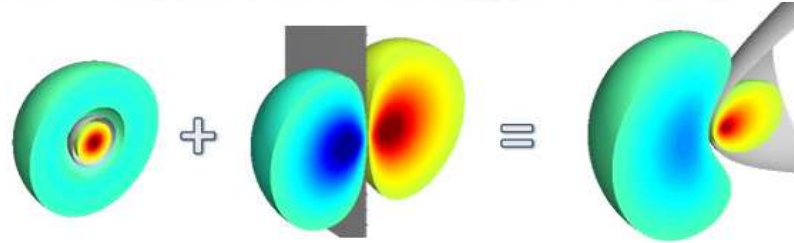
You can use ChemDraw for assistance!

Hybridization for MO

BeH_2

$$\begin{vmatrix} \varphi_1 \\ \varphi_2 \end{vmatrix} = \begin{vmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{vmatrix} \begin{vmatrix} s \\ p_z \end{vmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} \begin{vmatrix} s \\ p_z \end{vmatrix}$$

$Be: (1s)^2(\chi_1)^1(\chi_2)^1$



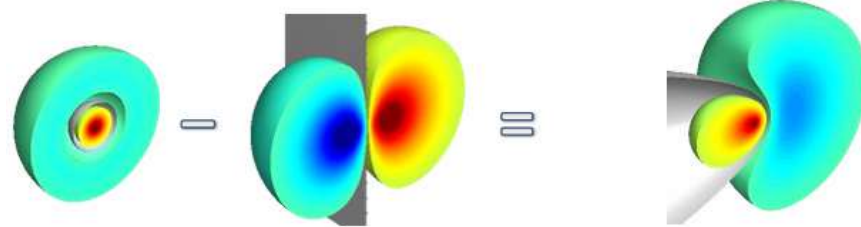
$$\chi_1(r) = \frac{1}{\sqrt{2}} [2s + 2p_z]$$

$$\frac{1}{\sqrt{2}} s$$

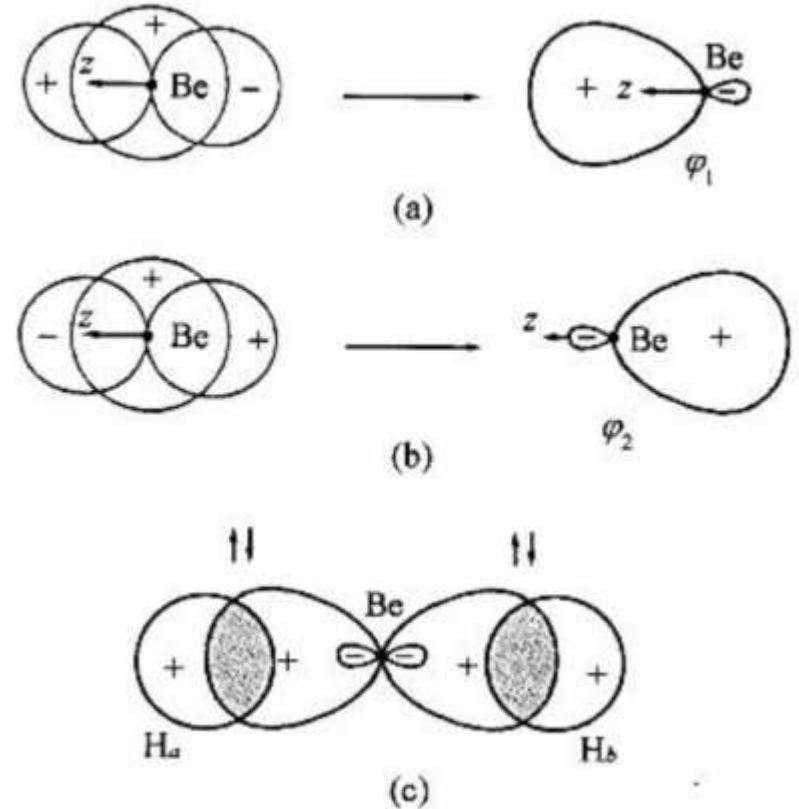
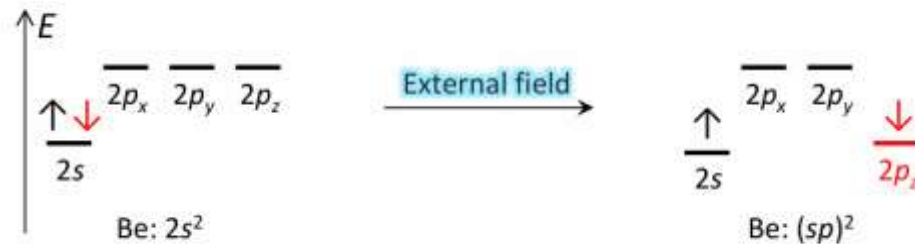
$$\frac{1}{\sqrt{2}} p_z$$

sp hybrid

$$\chi_2(r) = \frac{1}{\sqrt{2}} [2s - 2p_z]$$

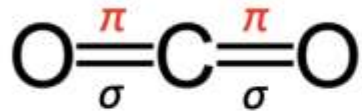
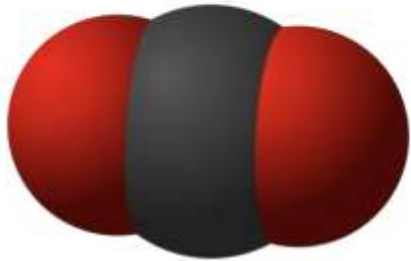


Normalization constant

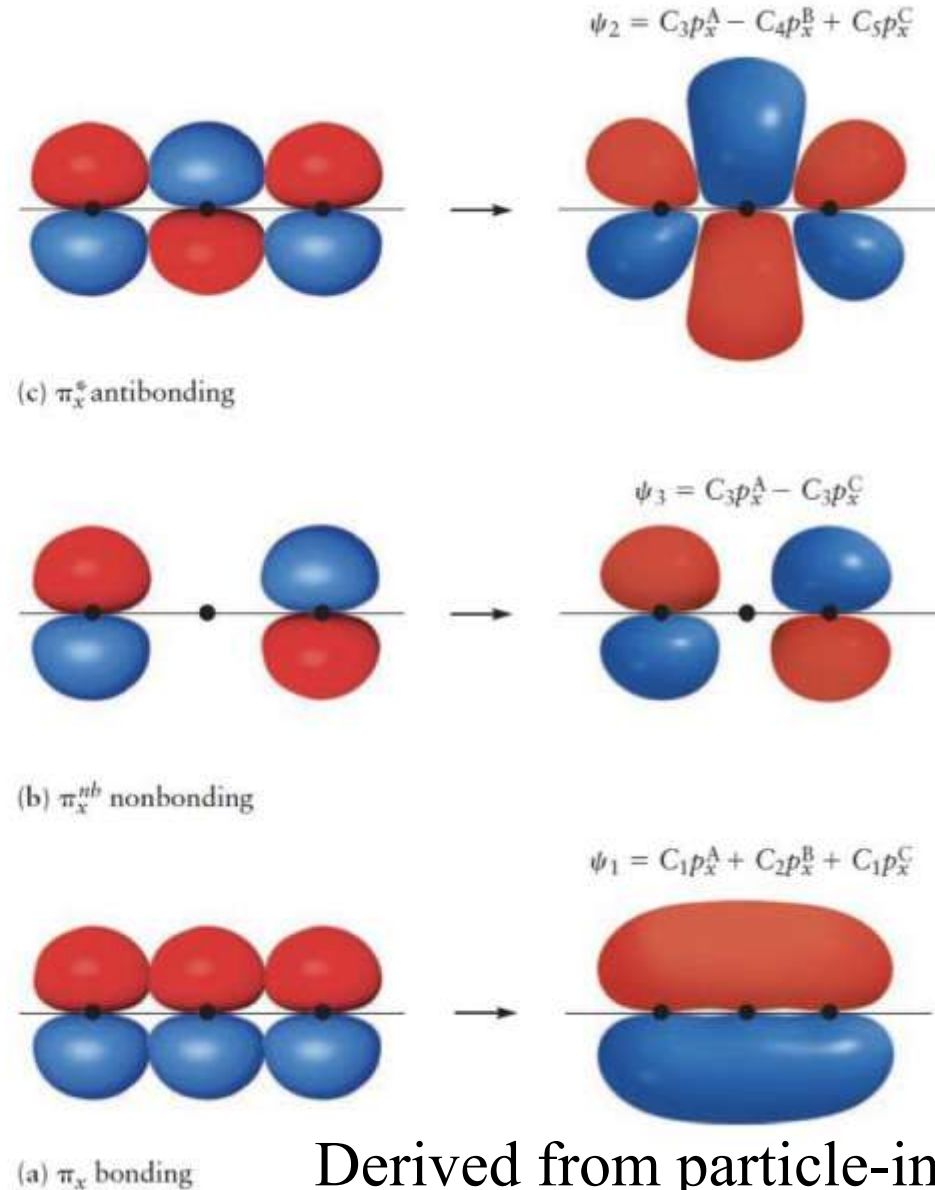


Hybridization for MO

Linear Triatomic Nonhydrides



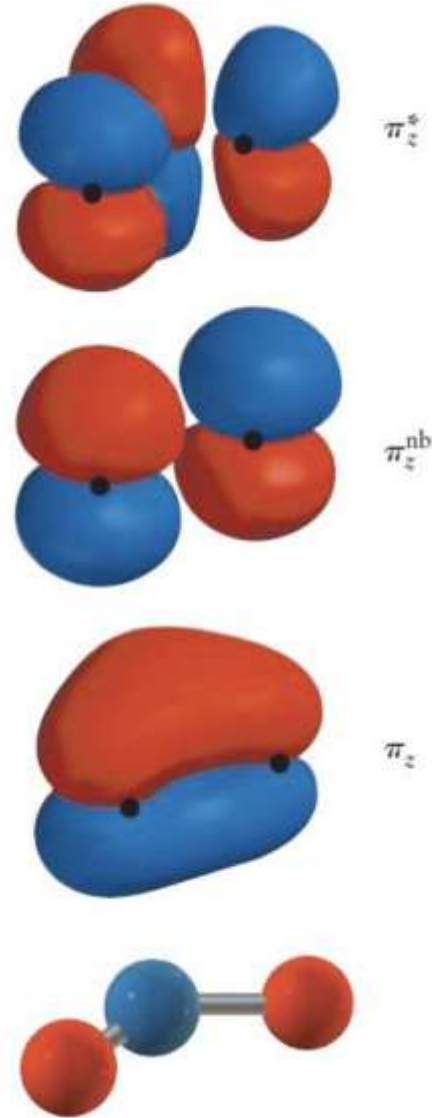
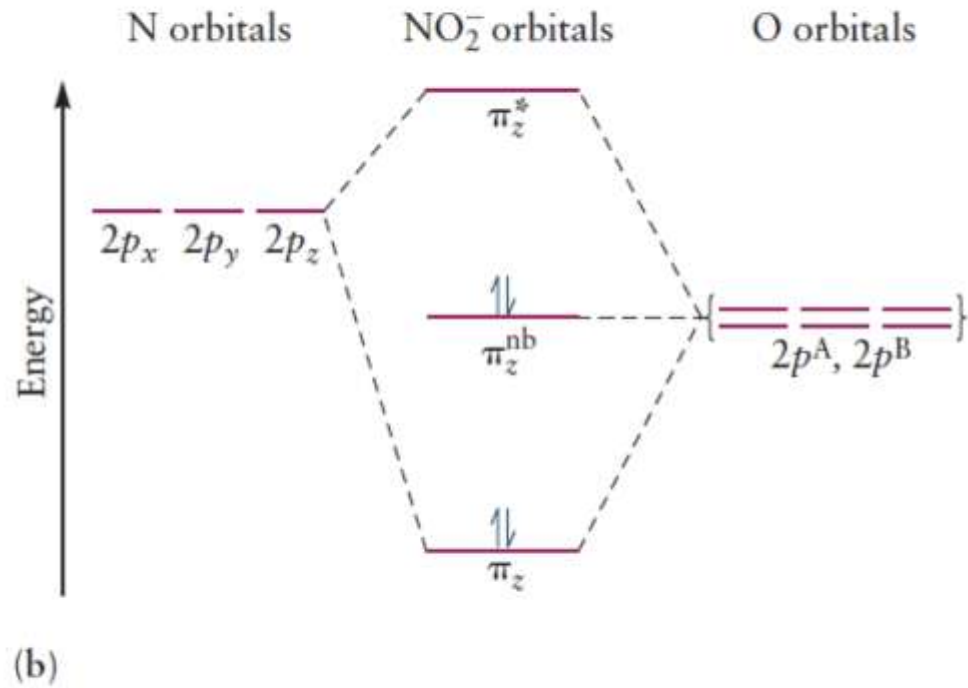
$$E_n = \frac{n^2 h^2}{8mL^2} \quad n = 1, 2, 3, \dots$$



Derived from particle-in-a-box model !

Hybridization for MO

Nonlinear Triatomic Molecules

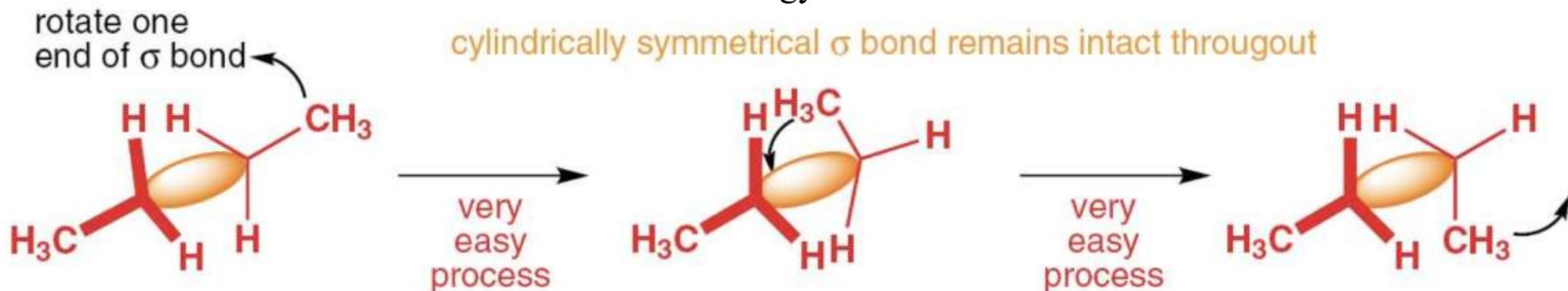


MO: Regions in a molecule where an electron occupying that orbital is likely to be found

Intro to organic chemistry

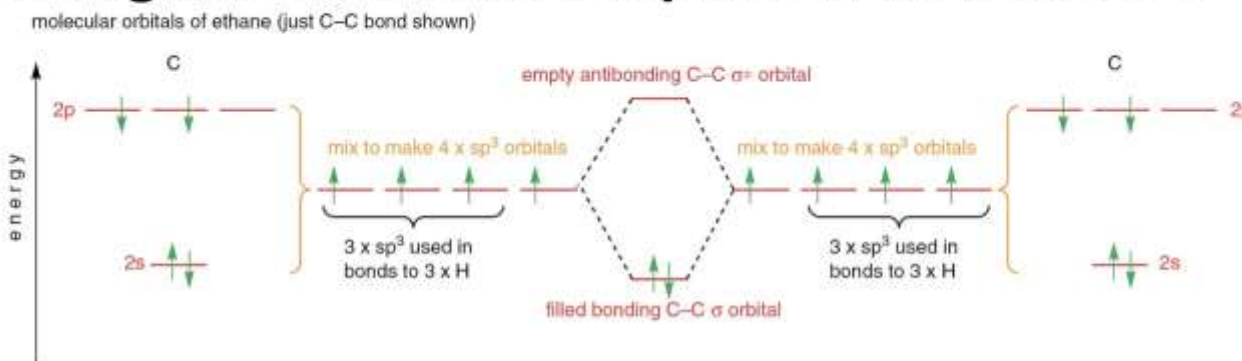
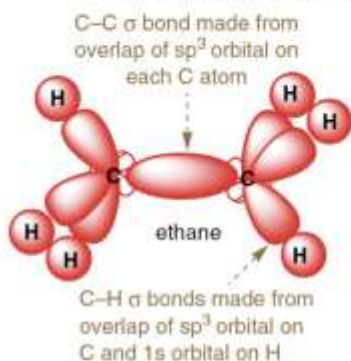
Conformation and bond rotate

Alkanes rotate freely...



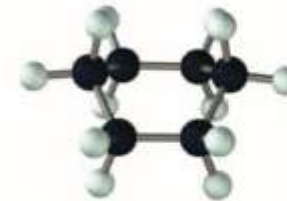
- sp^3 hybridization

- framework of organic molecules: replace H with more C



chair conformation

椅型构象



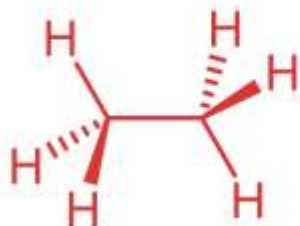
boat conformation

船型构象

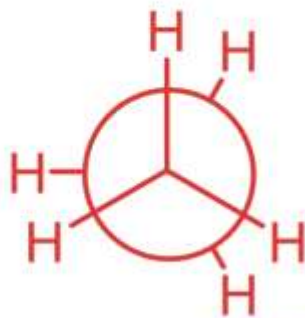
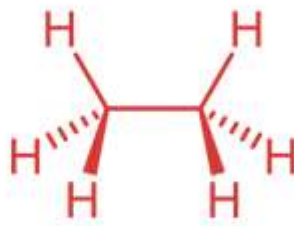
- Conformation is a fundamental concept in organic chemistry
- different conformations are invariably associated with variations in energy.

Intro to organic chemistry

Conformation and bond rotate

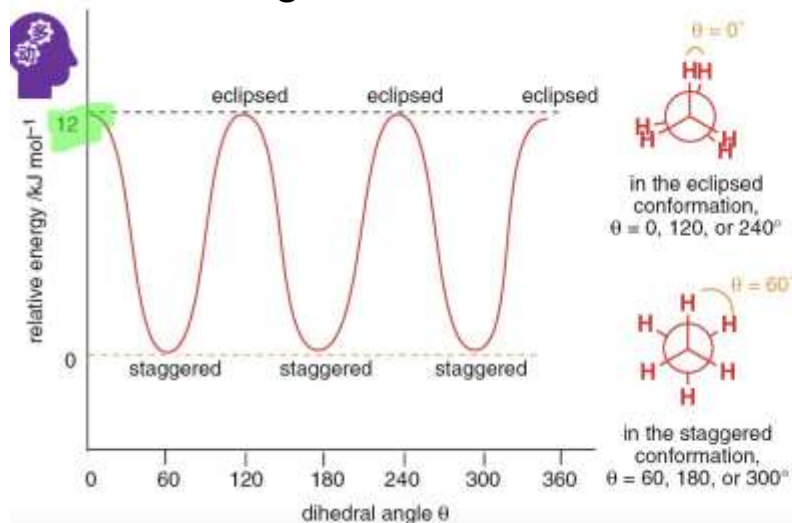


Staggered, 交叉型



Eclipsed, 重叠型

Dihedral angle 二面角



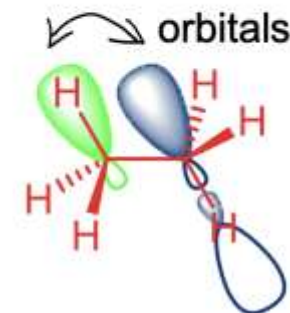
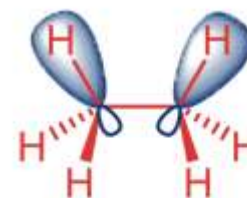
higher energy

anything in between: skew conformation



lower energy

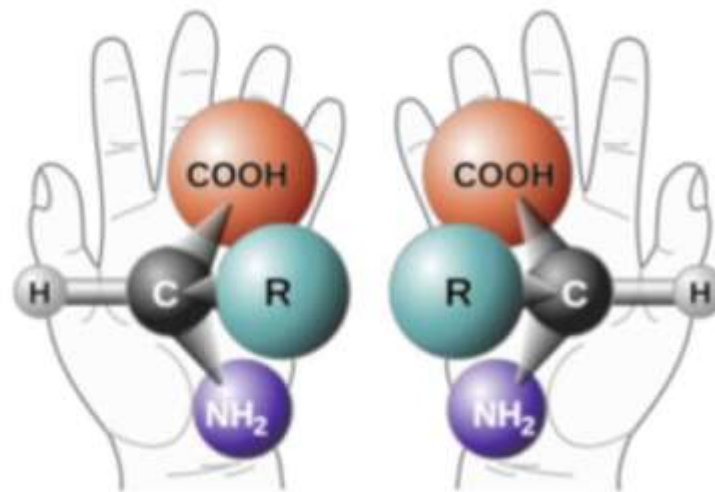
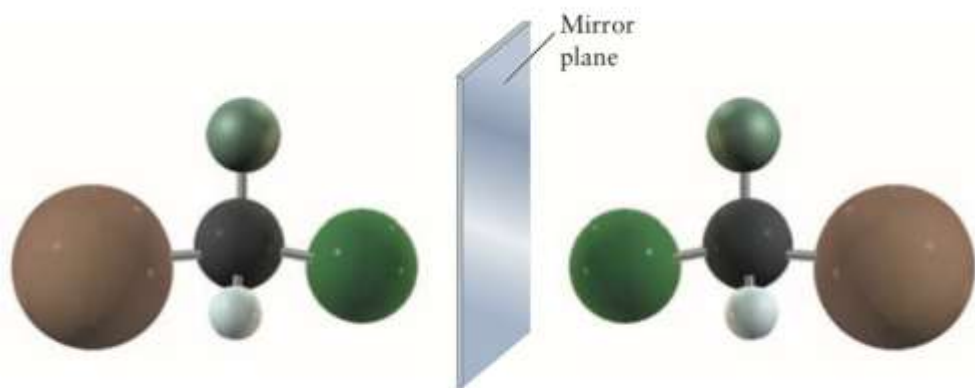
@ T = 298 K, $\Delta E = 12 \text{ kJ/mol}$
high e / low e ratio ~ 0.0079



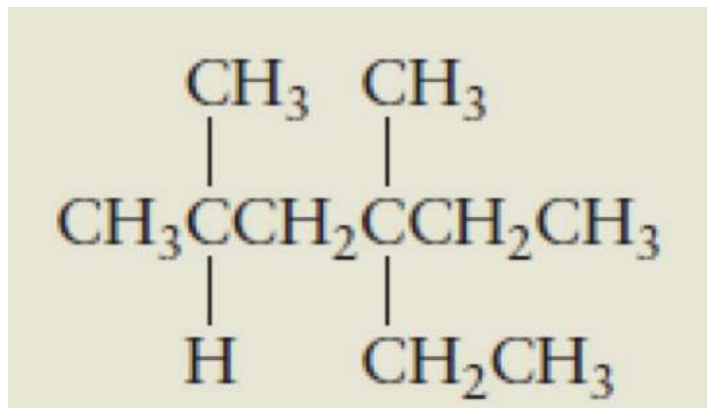
Electrons hate
electrons but love
empty orbitals

Intro to organic chemistry

Chiral and chirality



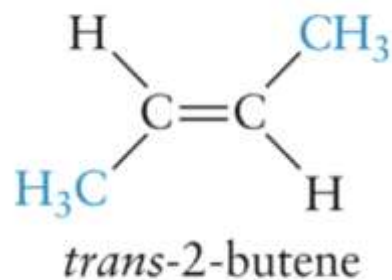
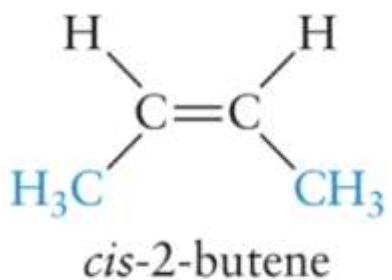
A molecule that lacks a plane of symmetry, a center of symmetry, and a four-fold alternating axis is chiral. Chiral molecules exhibit distinct optical properties.



Intro to organic chemistry



cis-2-butene



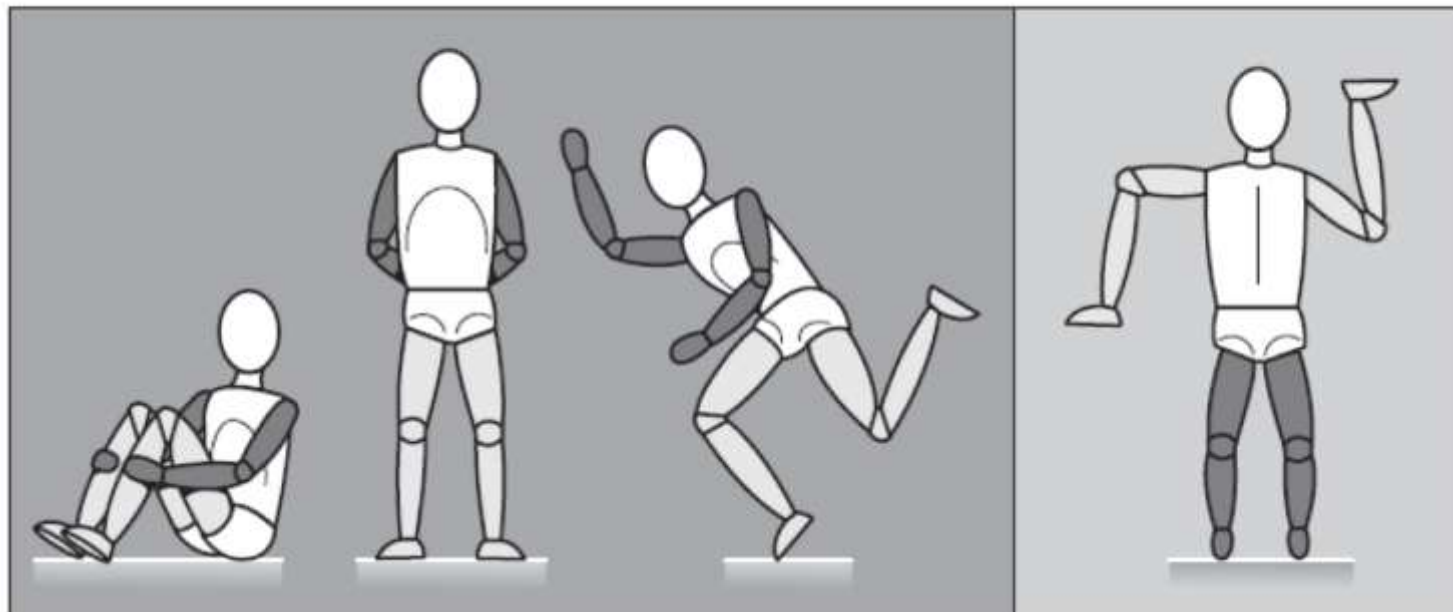
- **Breaking a π bond costs a significant amount of energy:**
both *cis* (顺式) and *trans* (反式) forms are stable, and
interconversion between the two is slow.

Double bond cannot rotate freely

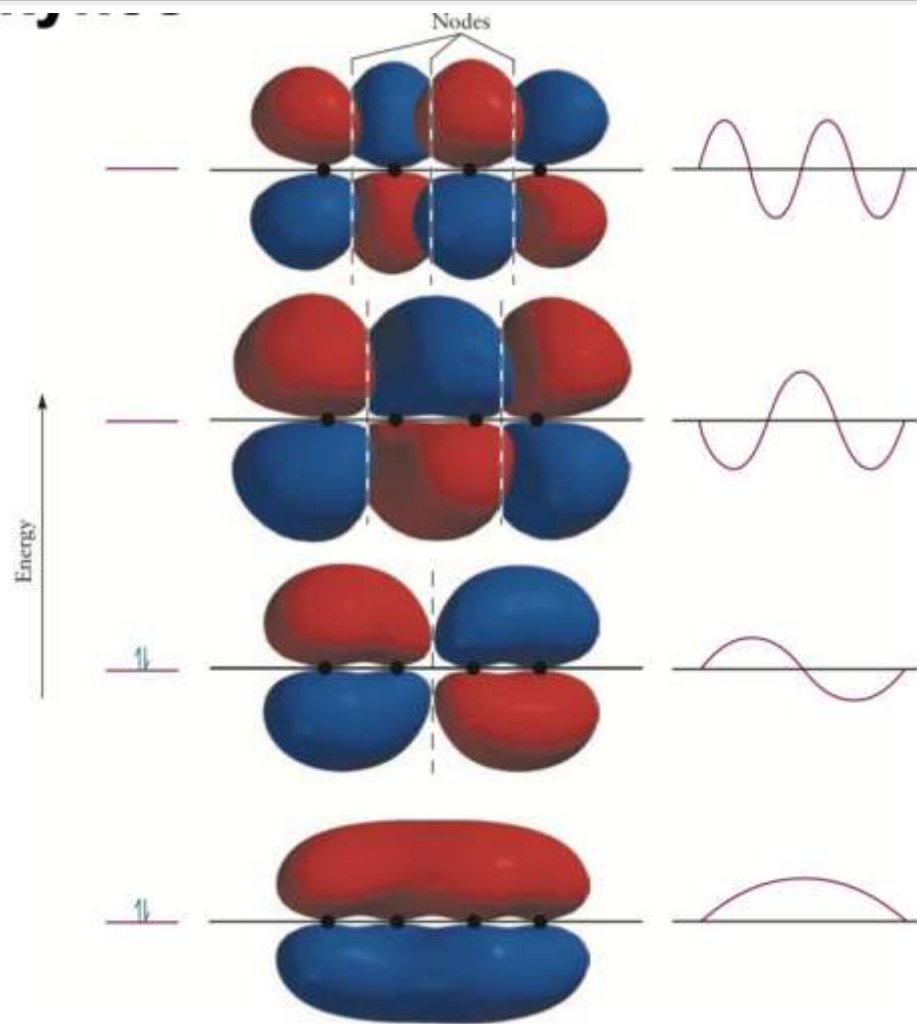
Interconversion among conformations
will not break any bond
Interconversion among configurations
(构型) **will** break bonds

Different conformations of a person – Some more stable than others ...

A different configuration

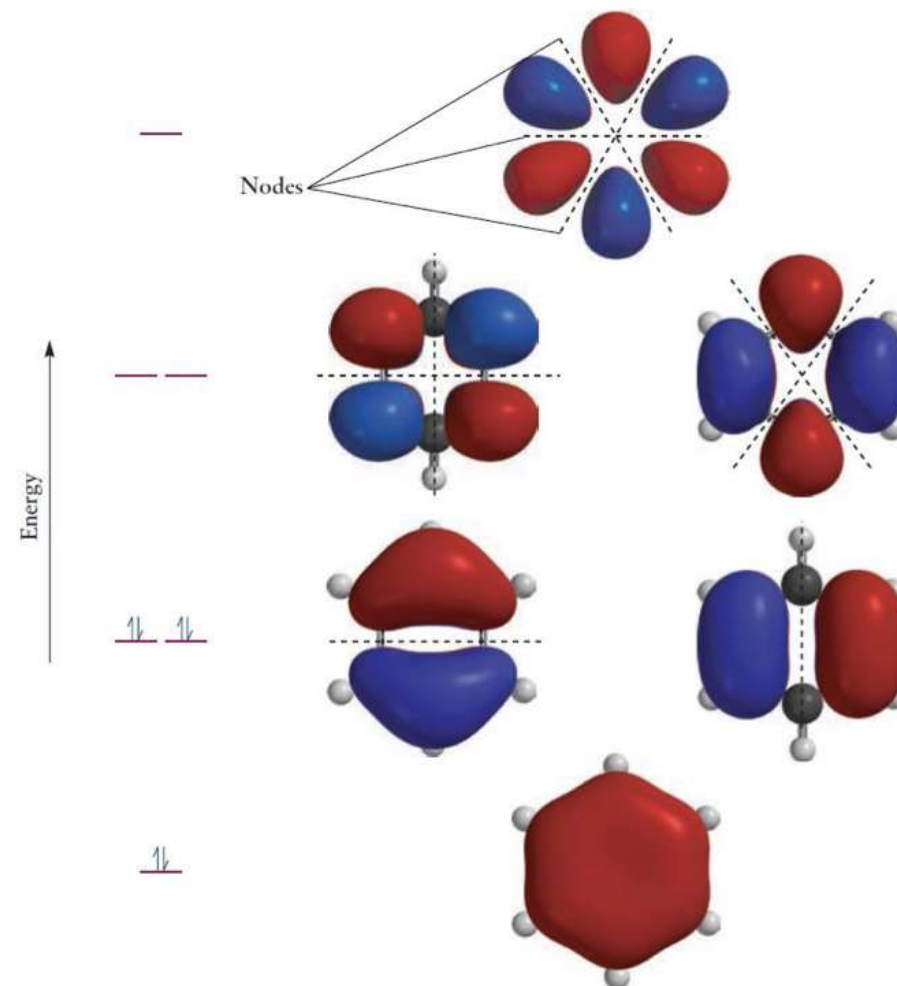


Frontier orbital theory and aromatics



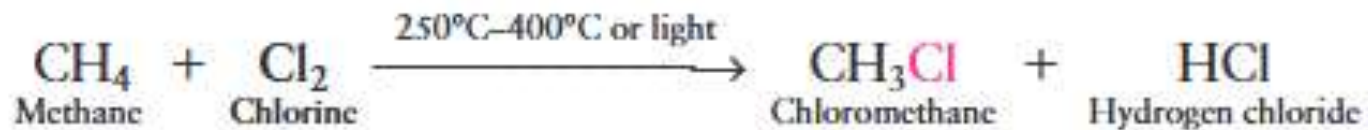
one-dimensional particle-in-a-box

The “ $4n+2$ ” rule



Free radical

A free radical is a chemical species that contains an odd (unpaired) electron; it is usually formed by breaking a covalent bond to form a pair of such species.



“链引发” “链转移” “链终止”